

Jaehyung Jung, Ph.D.

Postdoctoral researcher

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EXPERIENCES

- **Mobile Robotics Lab, Technical University of Munich** Sept. 2023 – Present
Postdoctoral Researcher (Supervisor: Prof. Stefan Leutenegger)
- **Republic of Korea Air Force** Jan. 2011 – Jan 2013
F-15K Fighter Jet Maintenance Staff Sergeant

EDUCATION

- **Ph.D. Aerospace Engineering, Seoul National University (SNU)** 2023
Thesis: Visual-Inertial Navigation System on Matrix Lie Group with Semantic Objects
- **M.S. Aerospace Engineering, Seoul National University** 2019
Thesis: Self-Calibrated Visual-Inertial Odometry for Rover Localization
- **B.S. Aerospace Engineering, Pusan National University** 2017
Magna Cum Laude (94%)

TECHNICAL SKILLS AND INTERESTS

Background: State estimation, Robot mapping, Machine learning for perception

Programming: C++, Python, ROS/ROS2, PyTorch, git

Languages: English (advanced), German (beginner), Korean (native)

KEY ACHIEVEMENTS IN RESEARCH PROJECTS

- **AUTOASSESS** 2023 – Present
Research Associate at TUM funded by Horizon Europe
 - Multi-sensor (camera/LiDAR/IMU/GNSS) fusion for platform-agnostic real-time state estimation [1, 2]
- **Indoor and outdoor integrated navigation technology** 2020 – 2023
Graduate Research Student at SNU funded by Korea Government
 - Object pose estimation with shape symmetry [4]
 - Uncertainty-aware visual-inertial state estimation [5]
 - Enhancing object pose estimation in an automotive scenario [7]
- **Pose estimation technology for mobile devices** 2018 – 2020
Graduate Research Student at SNU funded by Samsung Electronics
 - Event-inertial odometry for fast moving hand-held devices [8]
- **Integrated navigation system for lunar rover** 2017 – 2018
Graduate Research Student at SNU funded by Korea Government
 - Camera/IMU fusion for planetary rover localization, [YouTube](#)
- **2-DOF gimbal stabilizer in CANSAT competition** 2015 – 2016
Bachelor project
 - Automatic control of 2-DOF gimbal stabilizer in an embedded system, [YouTube](#)

OPENSOURCES

- **OKVIS2-X: Multi-sensor SLAM with volumetric dense mapping** in C++, Python, ROS2, PyTorch 2025
- **Gaussian Mixture Midway-Merge for Object SLAM** in MATLAB 2022
- **Ensemble Visual-Inertial-Odometry** in C++, ROS 2022

SELECTED PUBLICATIONS

• Co-authored 10+ publications (250+ citations) in leading robotics papers (ICRA, RA-L, T-RO)

• Please see [GoogleScholar](#) for the full list.

1. Simon Boche*, **Jaehyung Jung***, Sebastián Barbas Laina*, and Stefan Leutenegger (*equal contribution), “OKVIS2-X: Open Keyframe-based Visual-Inertial SLAM Configurable with Dense Depth or LiDAR, and GNSS,” *IEEE Transactions on Robotics*, vol. 41, pp. 6064-6083, 2025. [YouTube](#)
2. **Jaehyung Jung**, Simon Boche, Sebastián Barbas Laina, and Stefan Leutenegger, “Uncertainty-Aware Visual-Inertial SLAM with Volumetric Occupancy Mapping,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2025. [YouTube](#)
3. Sebastián Barbas Laina, Simon Boche, Sotiris Papatheodorou, Simon Schaefer, **Jaehyung Jung**, Stefan Leutenegger, “FindAnything: Open-Vocabulary and Object-Centric Mapping for Robot Exploration in Any Environment,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
4. **Jaehyung Jung** and Chan Gook Park, “Gaussian Mixture Midway-Merge for Object SLAM with Pose Ambiguity,” *IEEE Robotics and Automation Letters*, vol. 8, no. 1, pp. 400-407, 2023. [YouTube](#)
5. **Jaehyung Jung**, Yeongkwon Choe, and Chan Gook Park, “Photometric Visual-Inertial Navigation with Uncertainty-Aware Ensembles,” *IEEE Transactions on Robotics*, vol. 38, no. 4, pp. 2039-2052, 2022. [YouTube](#)
6. **Jaehyung Jung**, Jaehyuck Cha, Jae Young Chung, Tae Ihn Kim, Myung Hwan Seo, Sang Yeon Park, Jong Yun Yeo, and Chan Gook Park, “Monocular Visual-Inertial-Wheel Odometry using Low-Grade IMU in Urban Areas,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 2, pp. 925-938, 2022.
7. **Jaehyung Jung** and Chan Gook Park, “Object-based Visual-Inertial Navigation System on Matrix Lie Group,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2022. [YouTube](#)
8. **Jaehyung Jung** and Chan Gook Park, “Constrained Filtering-based Fusion of Images, Events, and Inertial Measurements for Pose Estimation,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2020. [YouTube](#)

ACADEMIC ACTIVITIES

- Co-organizing Workshop in ICRA 2025 2025
Novel Approaches for Precision Agriculture and Forestry with Autonomous Robots
- Organizing Workshop in RSS 2024 2024
Navigation & Mobile Manipulation in Challenging and Cluttered Natural Environments
- A reviewer in ICRA, IROS, RSS, RA-L, T-RO 2021 – Present

AWARDS

- Best Paper Award in Avionics Systems Symposium Korea 2019
- Best Paper Award in 33rd Institute of Control, Robotics, and Systems, Annual Conference 2018
- The 2nd Prize in CANSAT competition Korea 2016
 - Organized by Ministry of Science and ICT
 - Team “To the Space!”: 2-DOF camera gimbal stabilizer for CANSAT

TEACHING EXPERIENCES

- TA in Mobile Robotics (TUM) Theories in perception, control, and path planning 2024, 2023
- TA in Mobile Robotics Practicals (TUM) Autonomous flight (Point A-to-B with obstacles) 2024, 2023